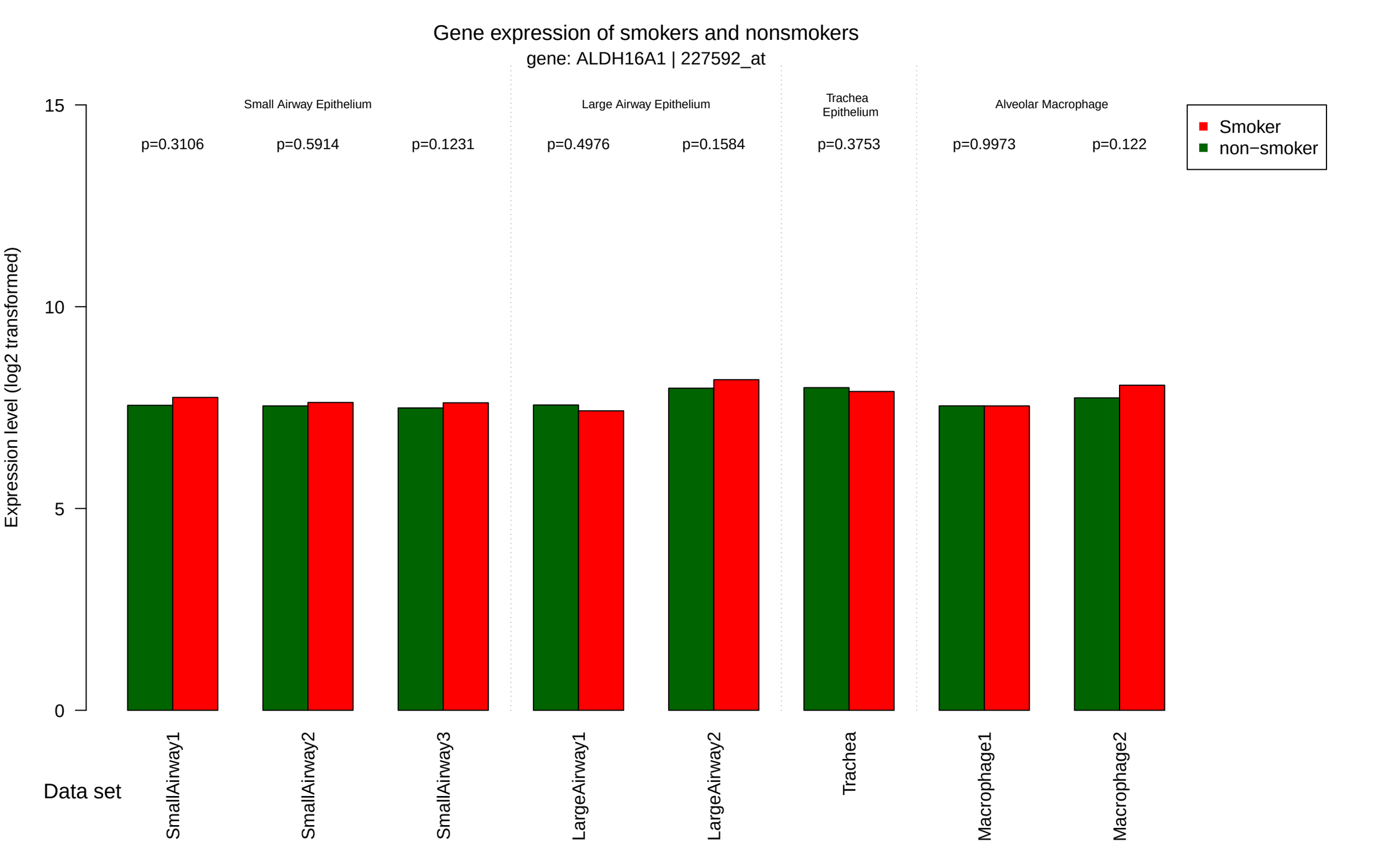
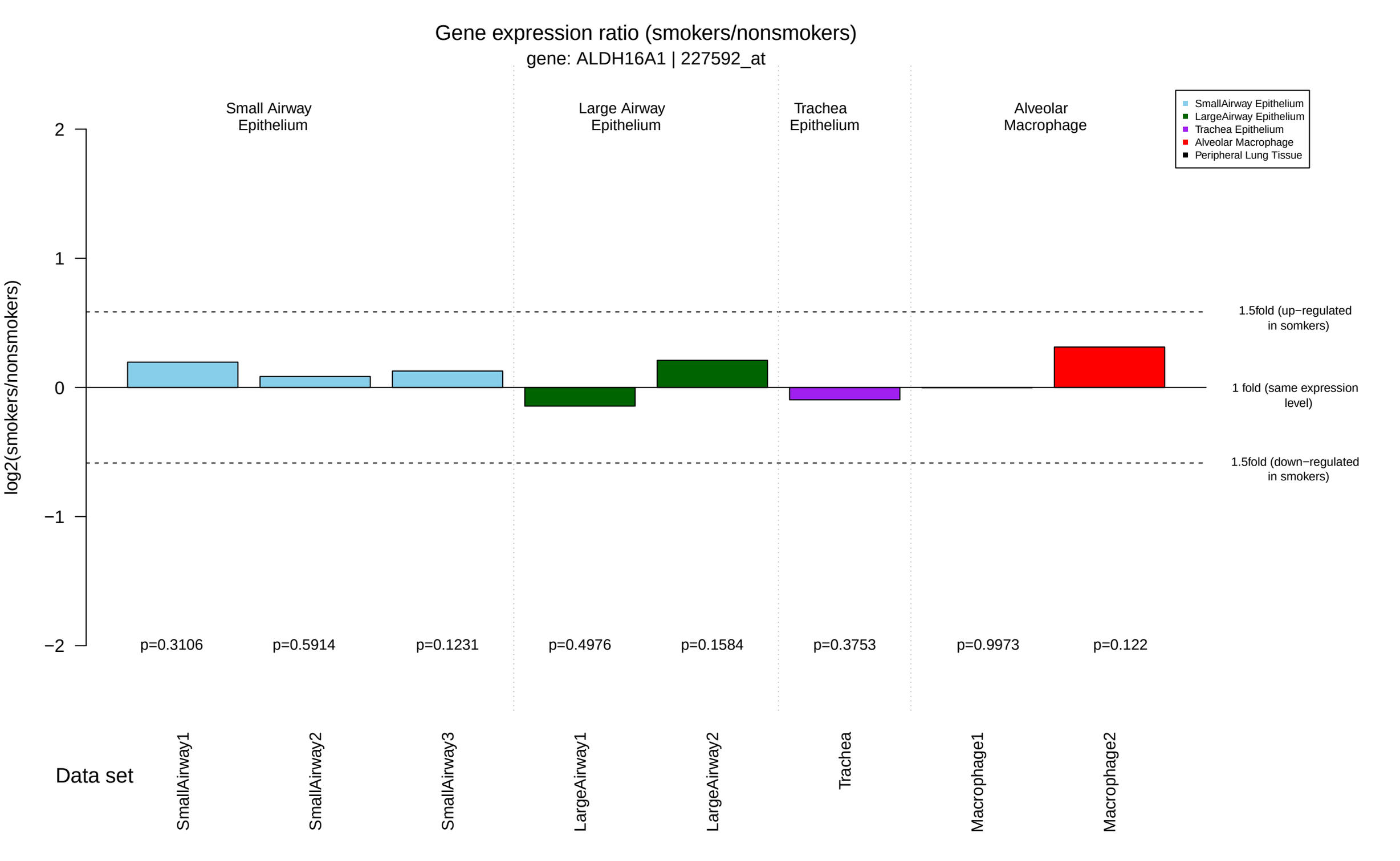
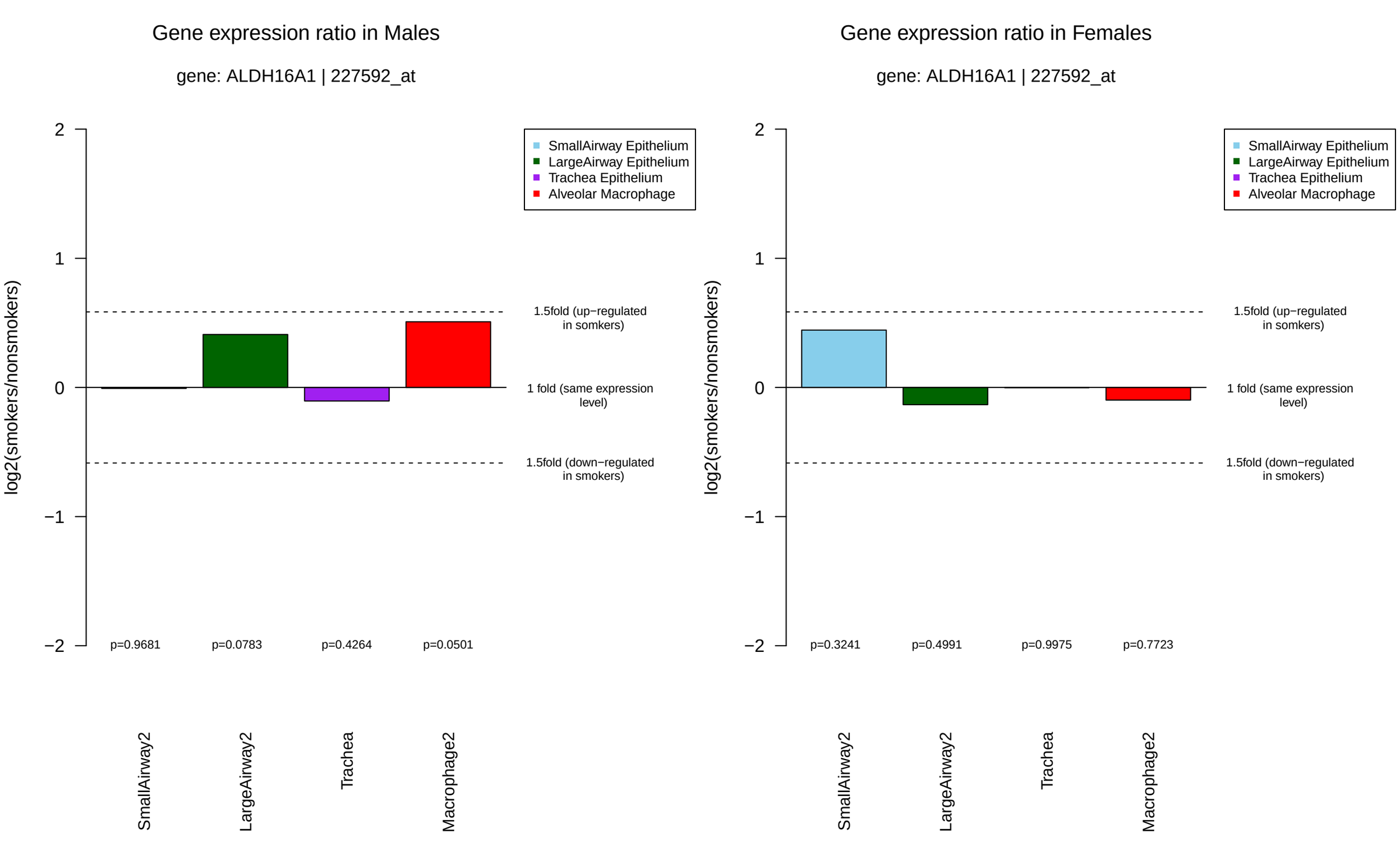
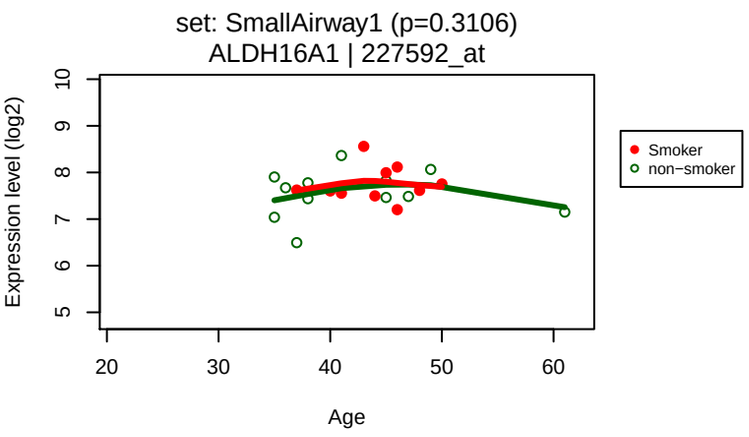
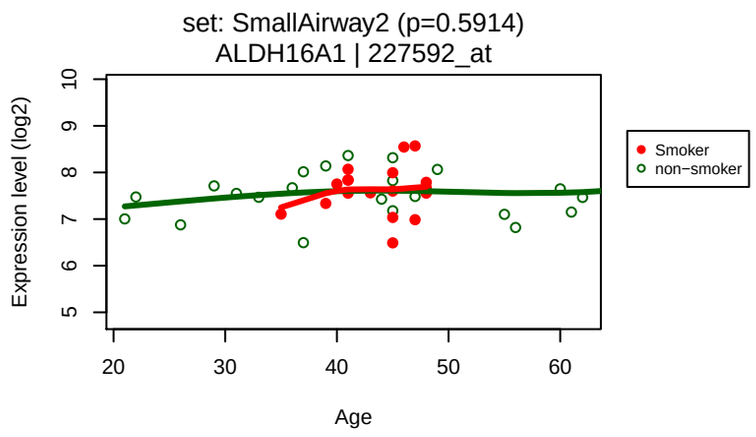
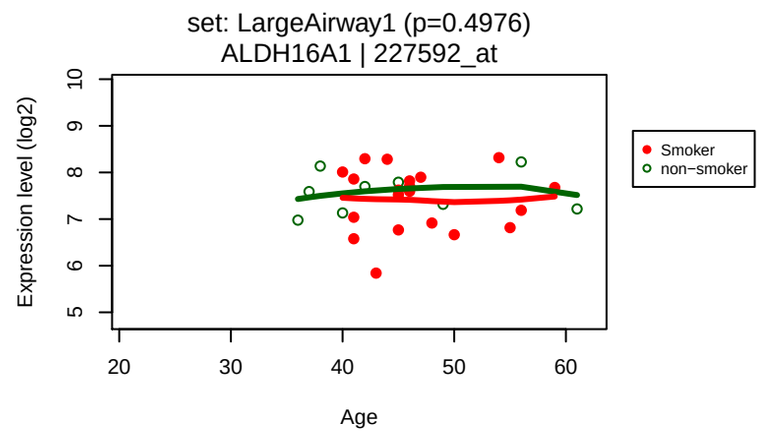

Gene expression of smokers and nonsmokers
gene: ALDH16A1 | 227592_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3106
p=0.5914
p=0.1231
p=0.4976
p=0.1584
p=0.3753
p=0.9973
p=0.122
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

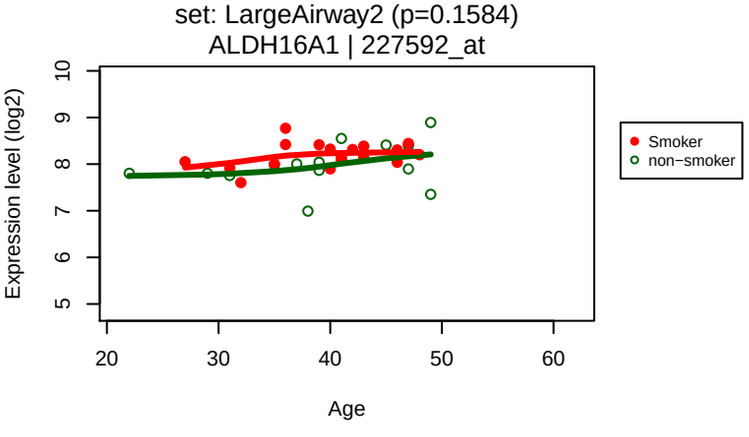

Gene expression ratio (smokers/nonsmokers)
gene: ALDH16A1 | 227592_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3106
p=0.5914
p=0.1231
p=0.4976
p=0.1584
p=0.3753
p=0.9973
p=0.122
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

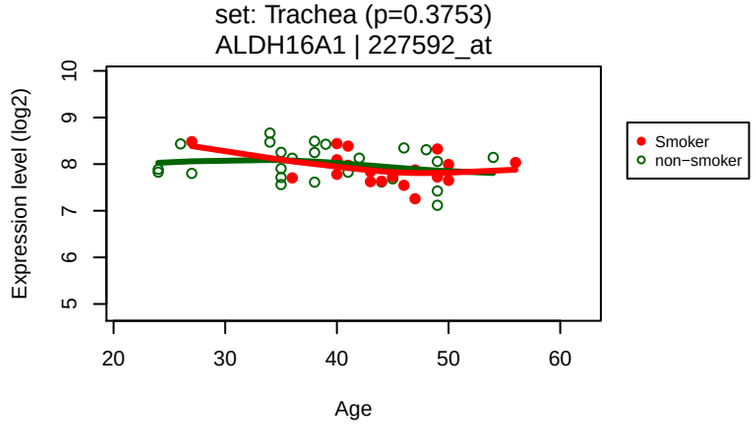

Gene expression ratio in Males
gene: ALDH16A1 | 227592_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9681
p=0.0783
p=0.4264
p=0.0501
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALDH16A1 | 227592_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3241
p=0.4991
p=0.9975
p=0.7723
SmallAirway2
LargeAirway2
Trachea
Macrophage2

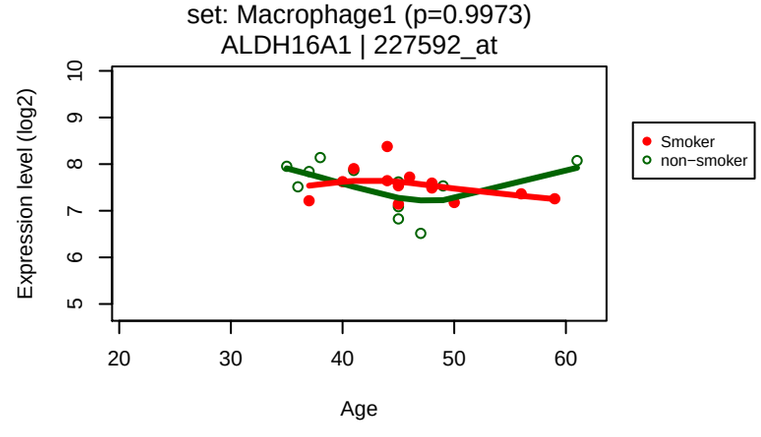

set: SmallAirway1 (p=0.3106)
ALDH16A1 | 227592_at
Expression level (log2)
Age
Smoker
non−smoker

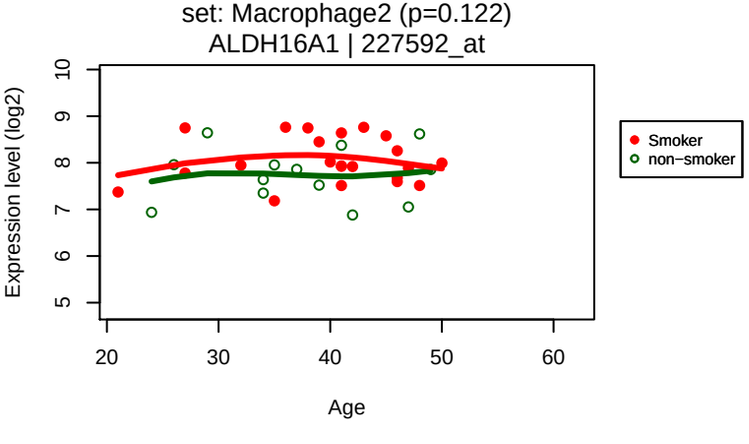

set: SmallAirway2 (p=0.5914)
ALDH16A1 | 227592_at
Expression level (log2)
Age
Smoker
non−smoker

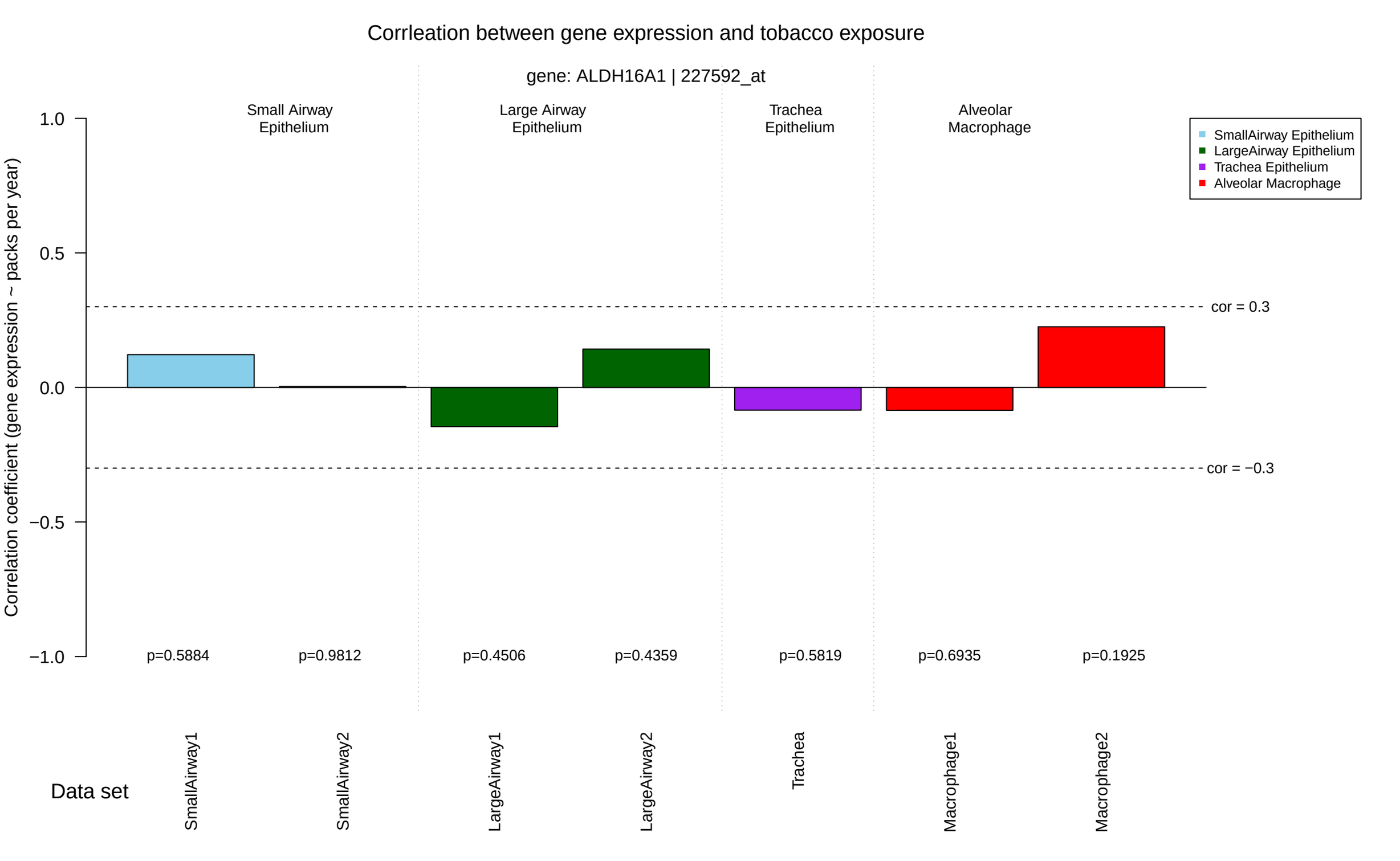

set: LargeAirway1 (p=0.4976)
ALDH16A1 | 227592_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1584)
ALDH16A1 | 227592_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3753)
ALDH16A1 | 227592_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9973)
ALDH16A1 | 227592_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.122)
ALDH16A1 | 227592_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALDH16A1 | 227592_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5884
p=0.9812
p=0.4506
p=0.4359
p=0.5819
p=0.6935
p=0.1925
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2